Name
Date
Course

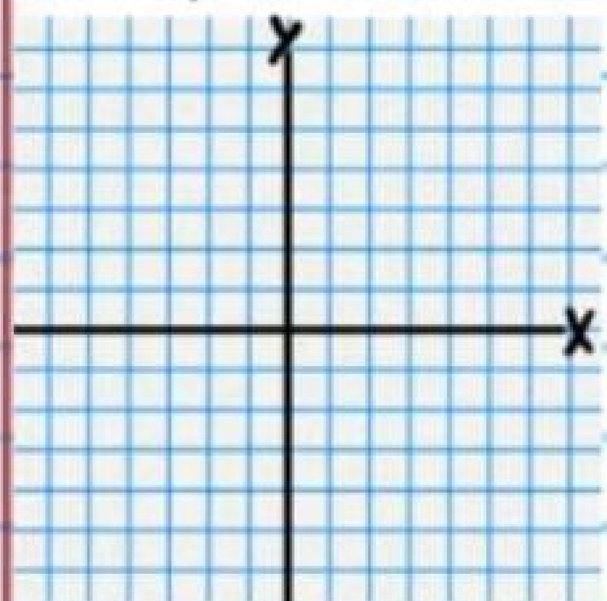
Transformations and the Library of Relations (Part III) Homework

Graph each equation below by performing multiple transformations on the parent graph. If a graph has asymptote(s), draw them as dotted lines.

① $\frac{y}{2}+3=\sqrt{-(x-4)}$

Parent Relation

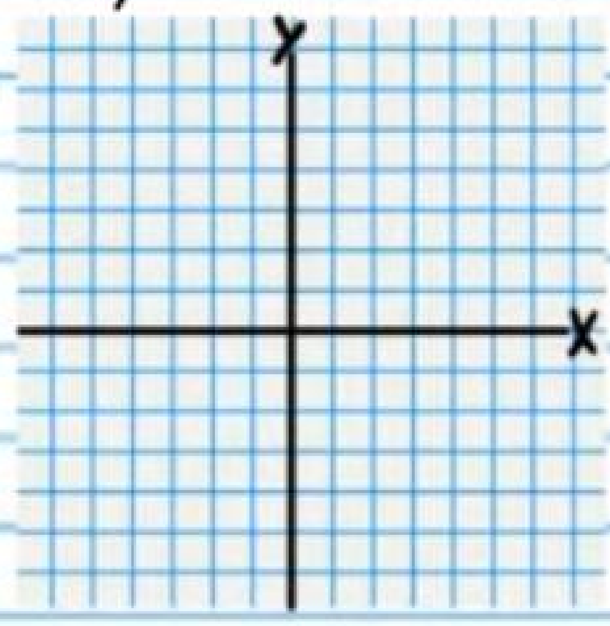
$y=\sqrt{x}$



Transformation #1

Reflection

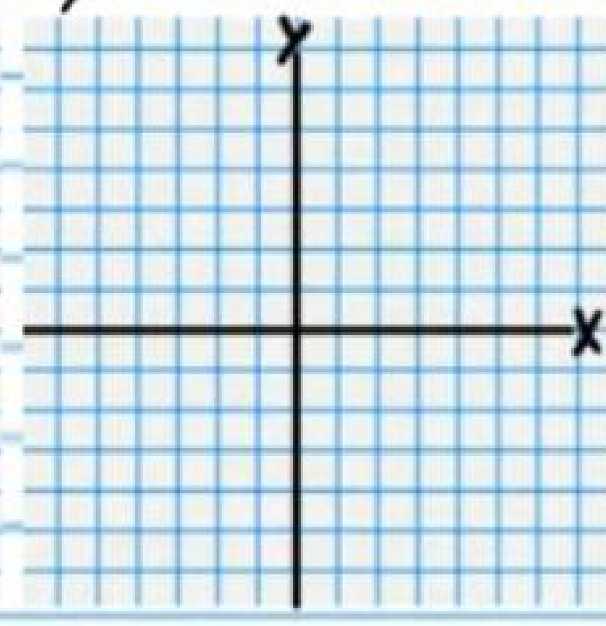
$y=\sqrt{-x}$



Transformation #2

Translation

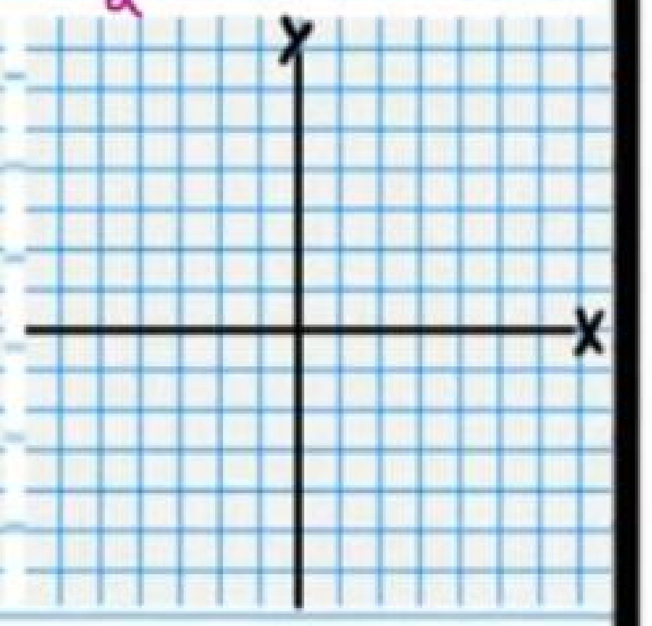
$y+3=\sqrt{-(x-4)}$



Final Transformation

Dilation

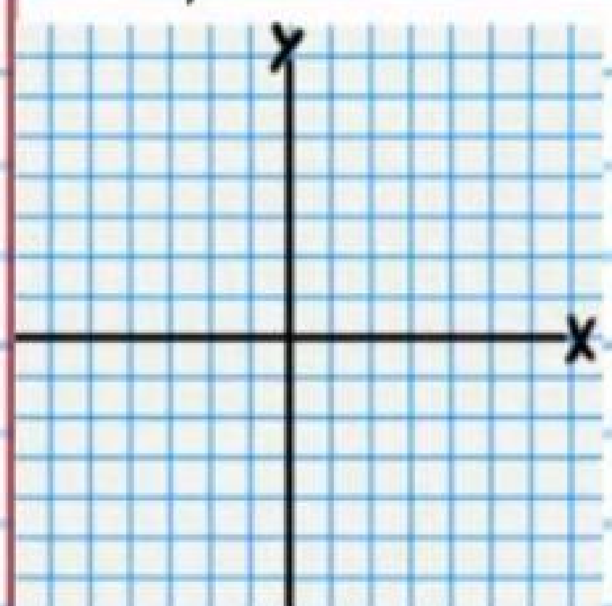
$\frac{y}{2}+3=\sqrt{-(x-4)}$



② $-y-1=(2x+3)^2$

Parent Relation

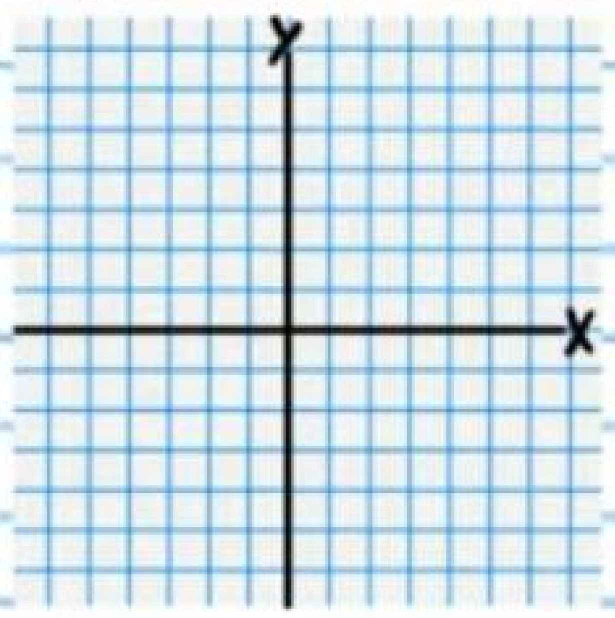
$y=x^2$



Transformation #1

Translation

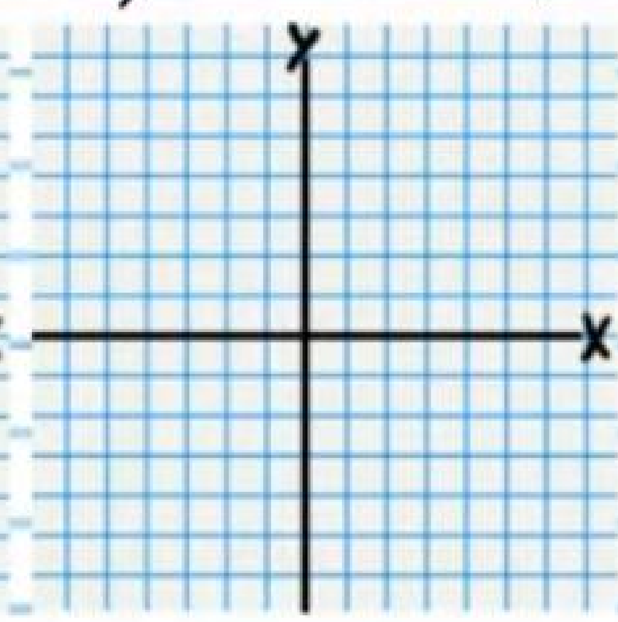
$y-1=(x+3)^2$



Transformation #2

Dilation

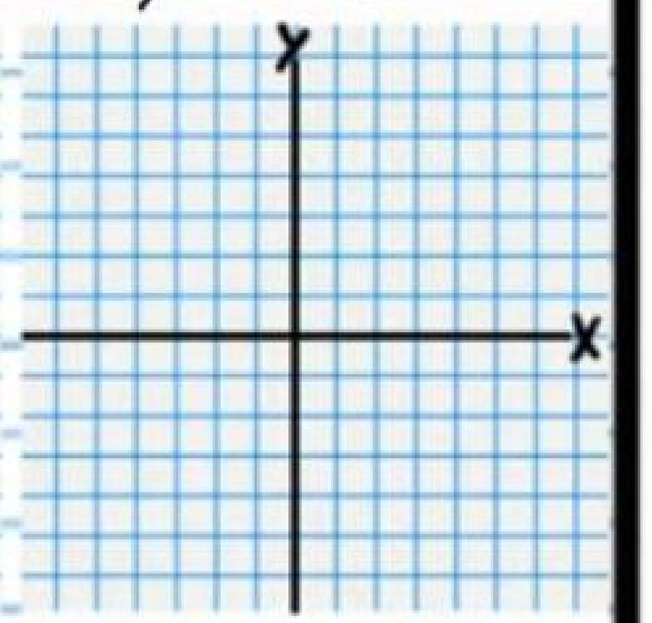
$y-1=(2x+3)^2$



Final Transformation

Reflection

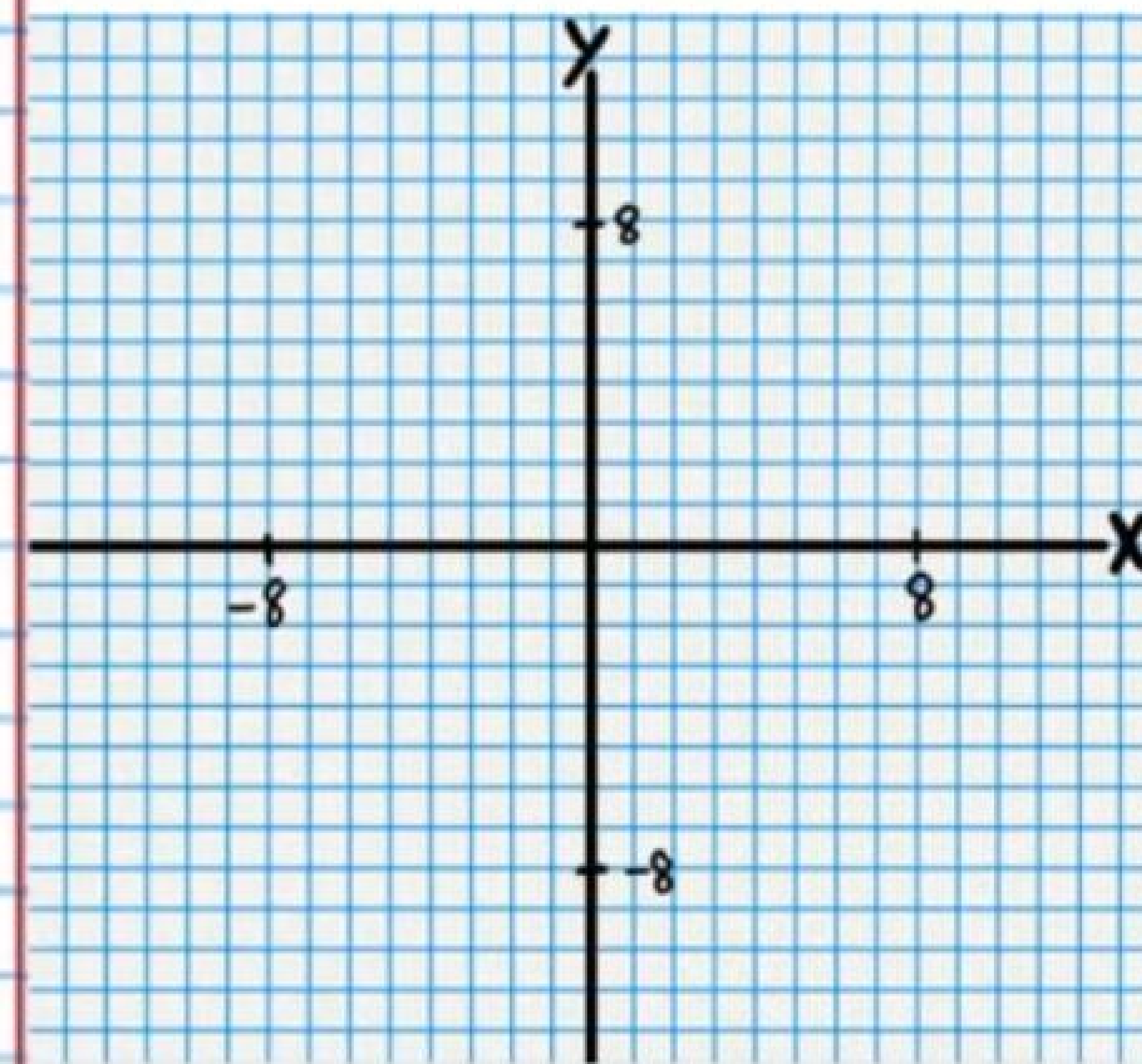
$-y-1=(2x+3)^2$



③ $\frac{-(y-2)}{3}=(x-2)^3$

Parent Relation

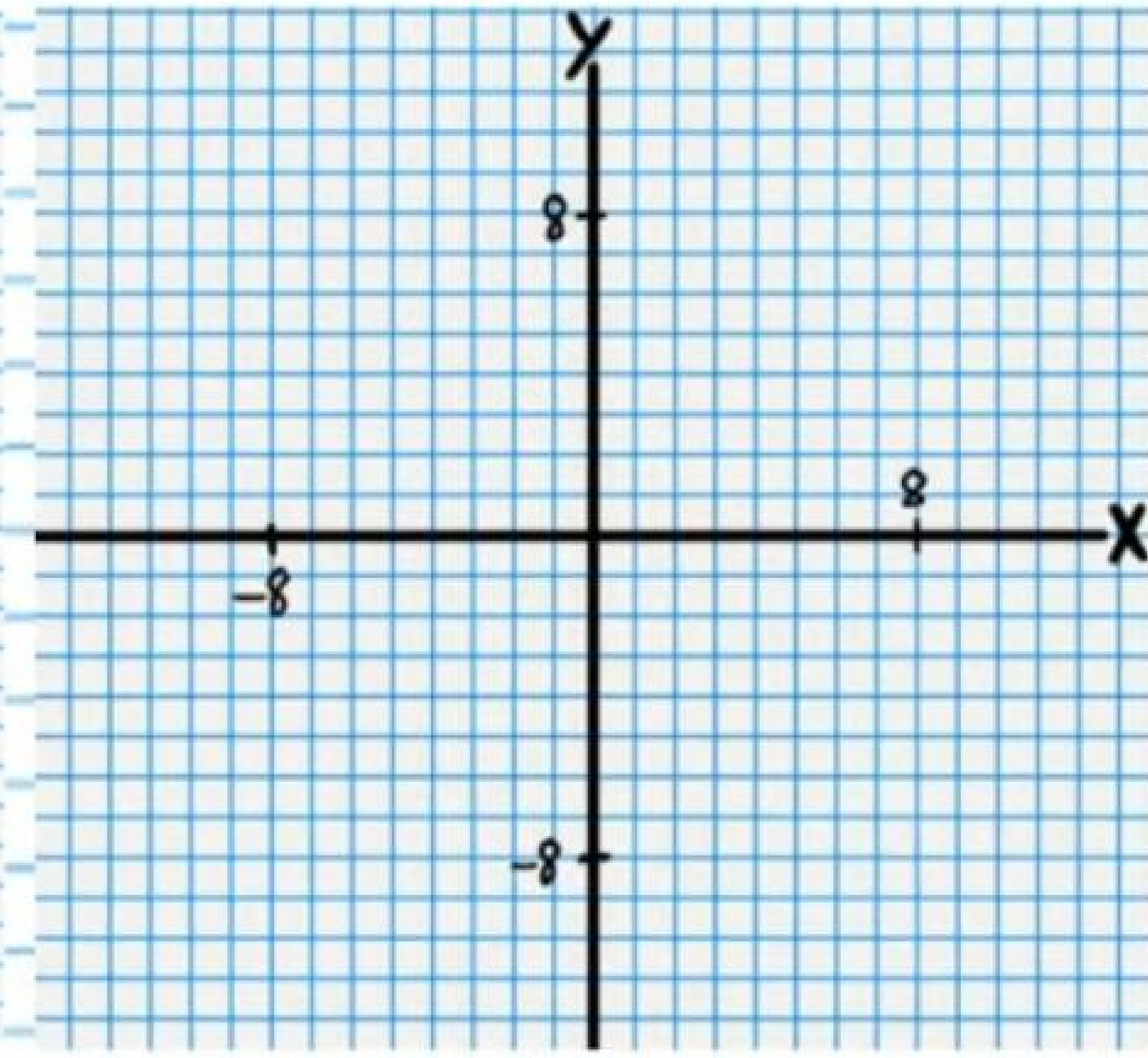
$y=x^3$



Transformation #1

Dilation

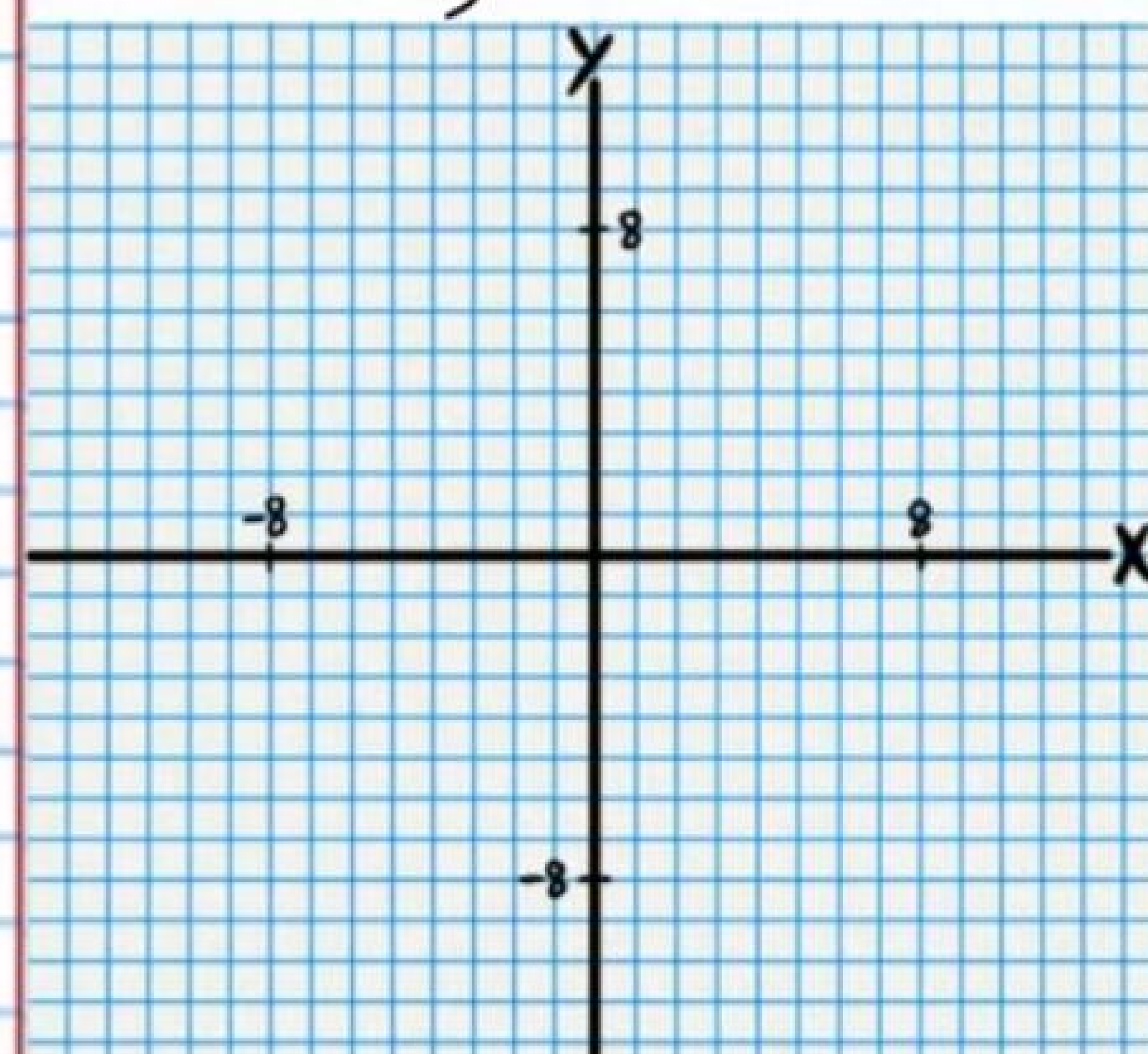
$\frac{y}{3}=x^3$



Transformation #2

Reflection

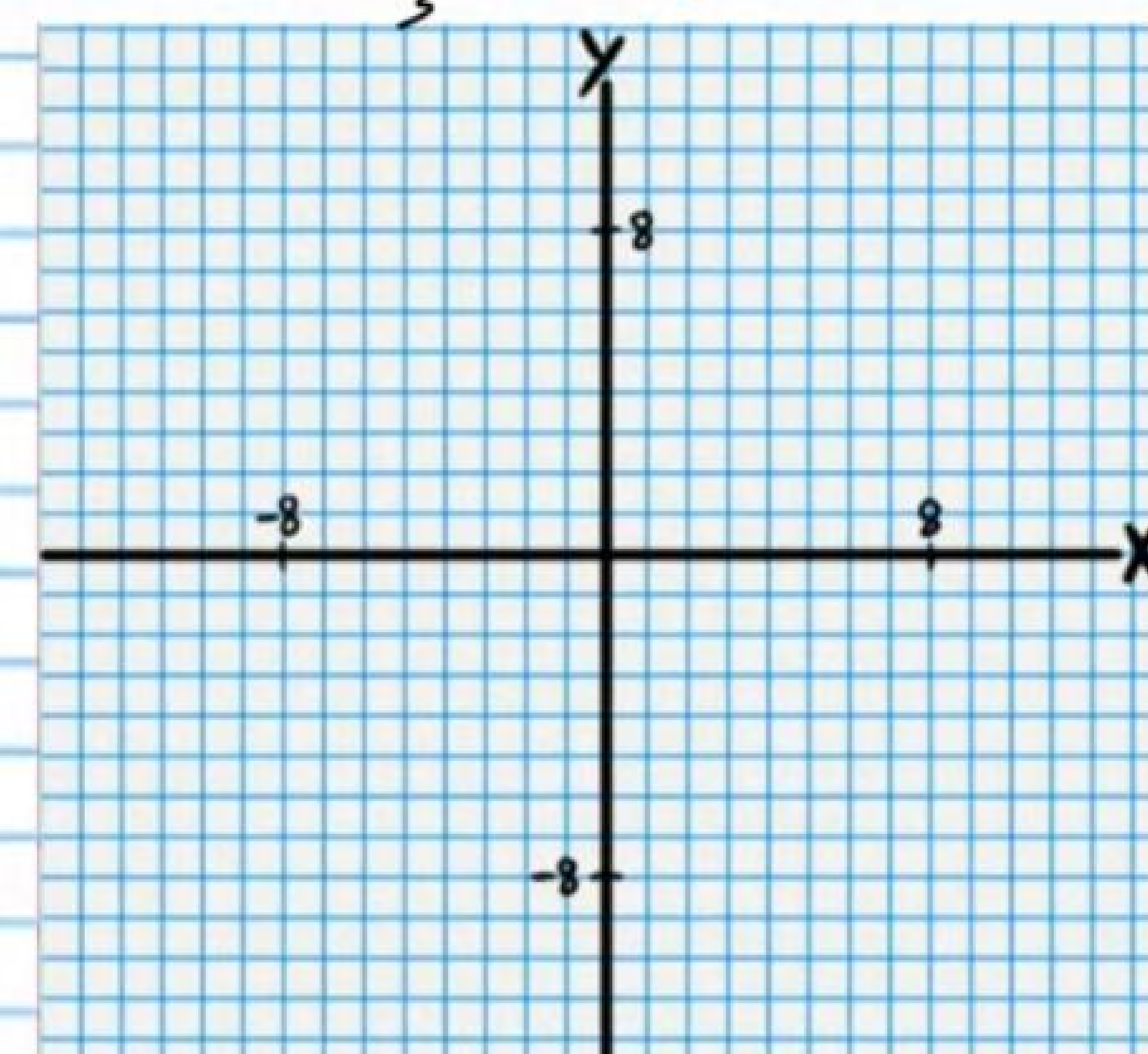
$-\frac{y}{3}=x^3$



Final Transformation

Translation

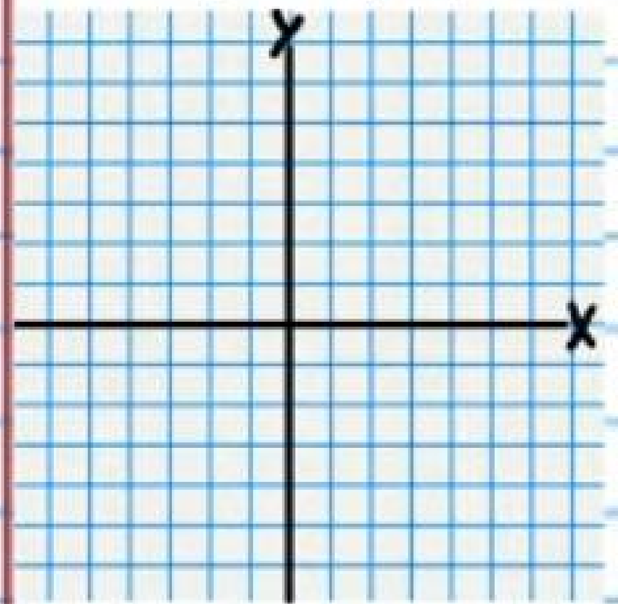
$\frac{-(y-2)}{3}=(x-2)^3$



(4) $-(y+1) = \left|\frac{x}{3}+1\right|$

Parent Relation

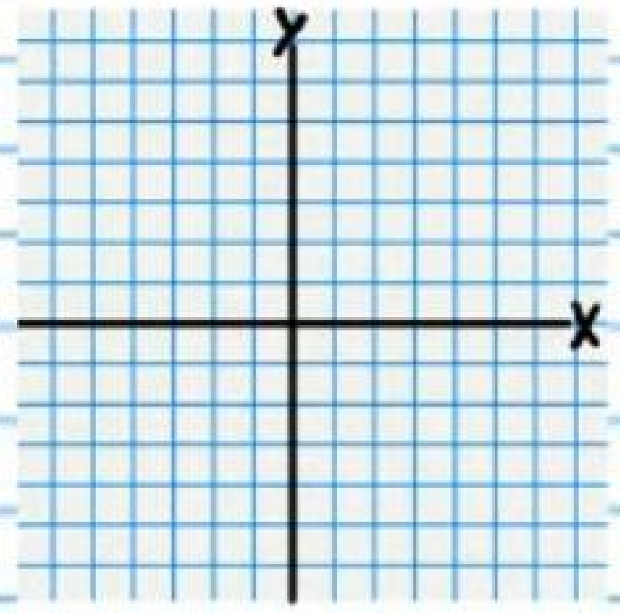
$y = |x|$



Transformation #1

Reflection

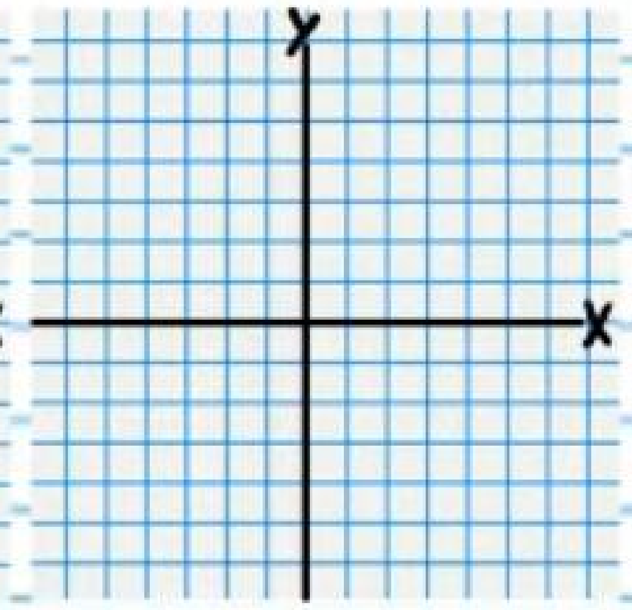
$-y = |x|$



Transformation #2

Translation

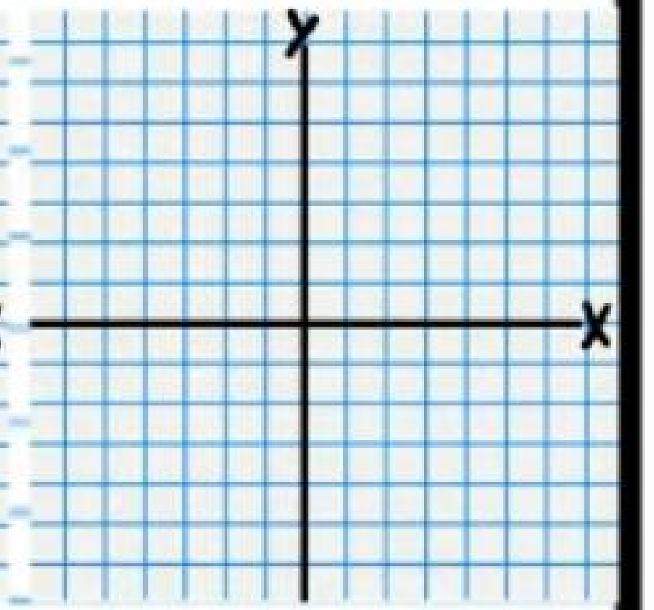
$-(y+1) = |x+1|$



Final Transformation

Dilation

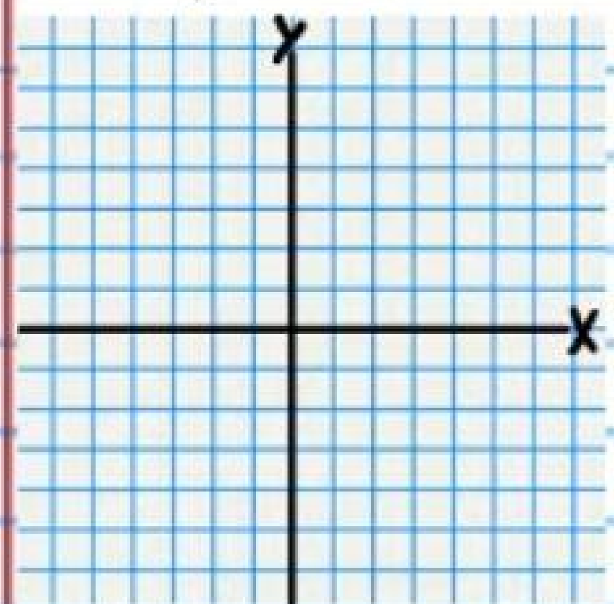
$-(y+1) = \left|\frac{x}{3}+1\right|$



(5) $2y+3 = \sqrt{-x+1}$

Parent Relation

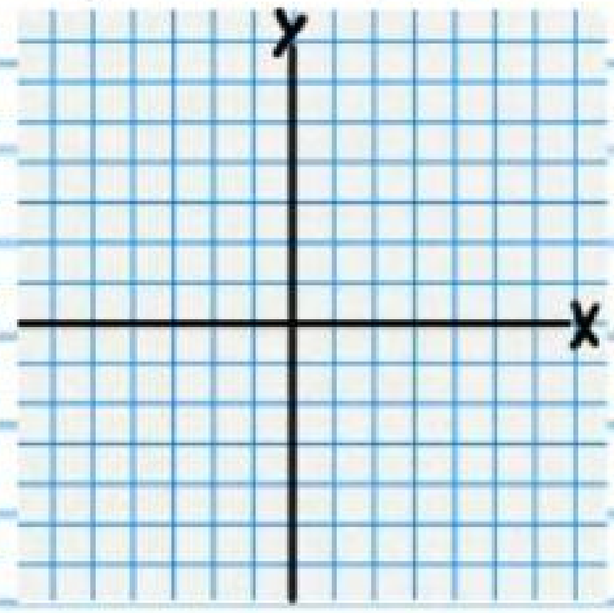
$y = \sqrt{x}$



Transformation #1

Translation

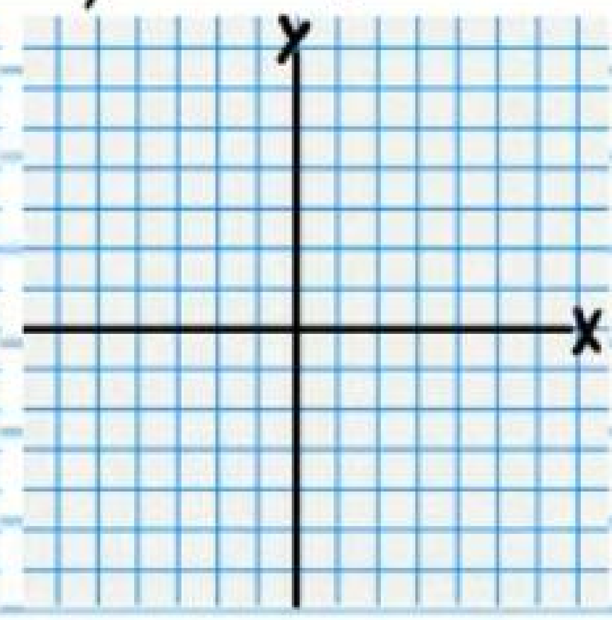
$y+3 = \sqrt{x+1}$



Transformation #2

Reflection

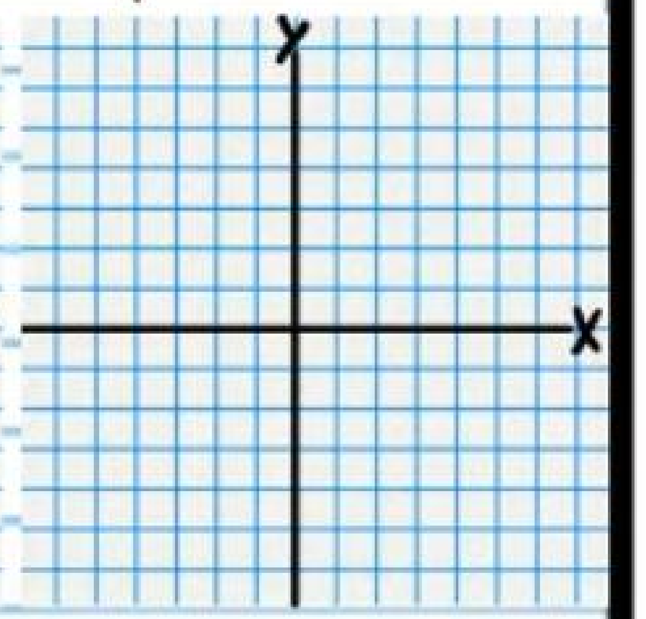
$y+3 = \sqrt{-x+1}$



Final Transformation

Dilation

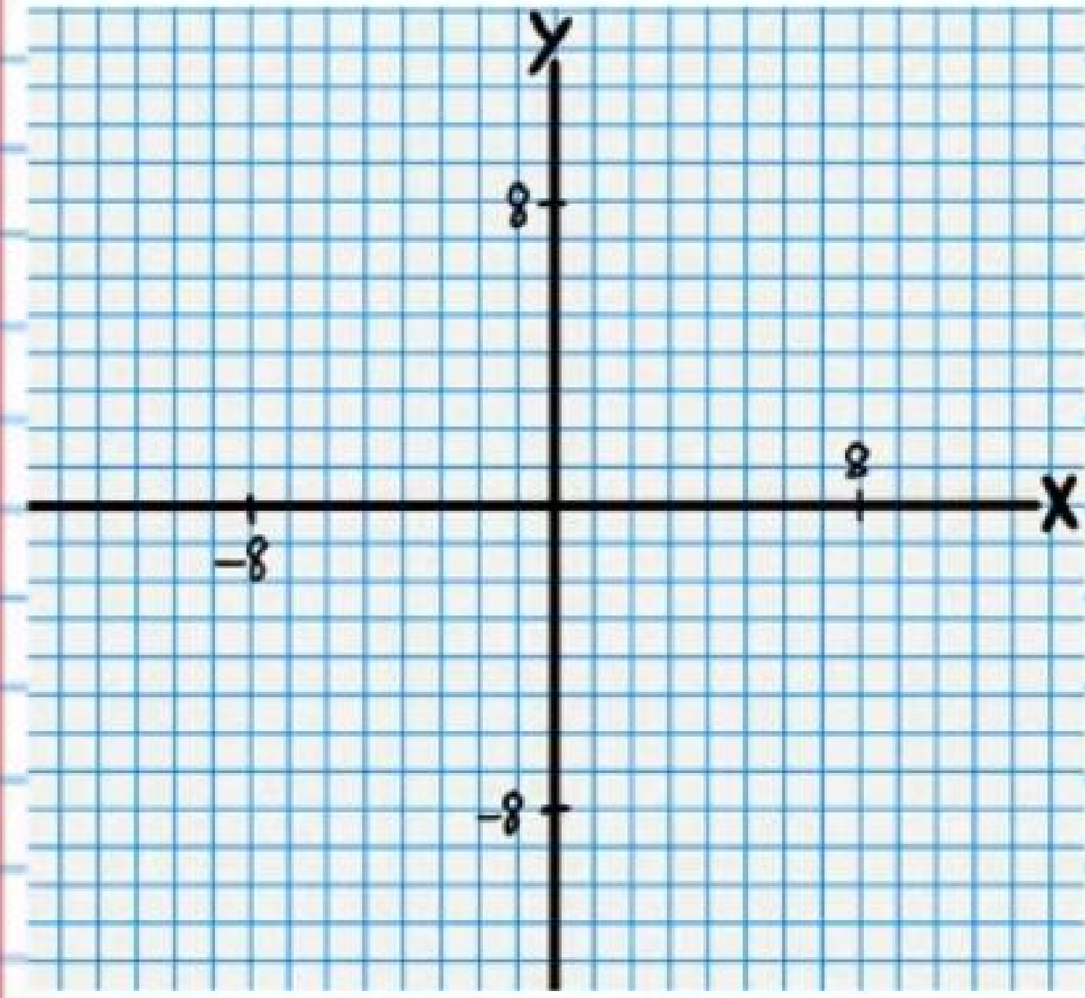
$2y+3 = \sqrt{-x+1}$



(6) $-(y-2) = 3^{\frac{x-1}{2}}$

Parent Relation

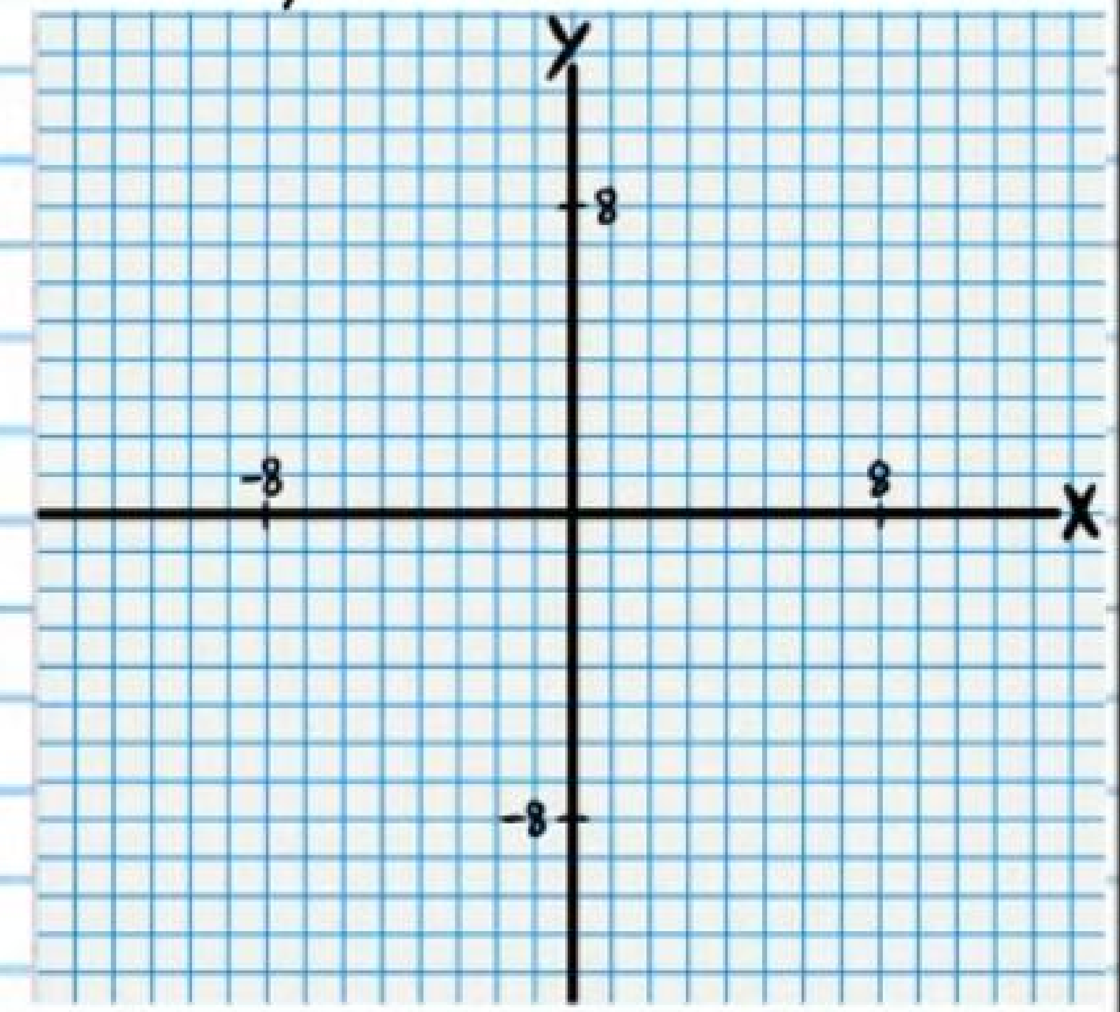
$y = 3^x$



Transformation #1

Reflection

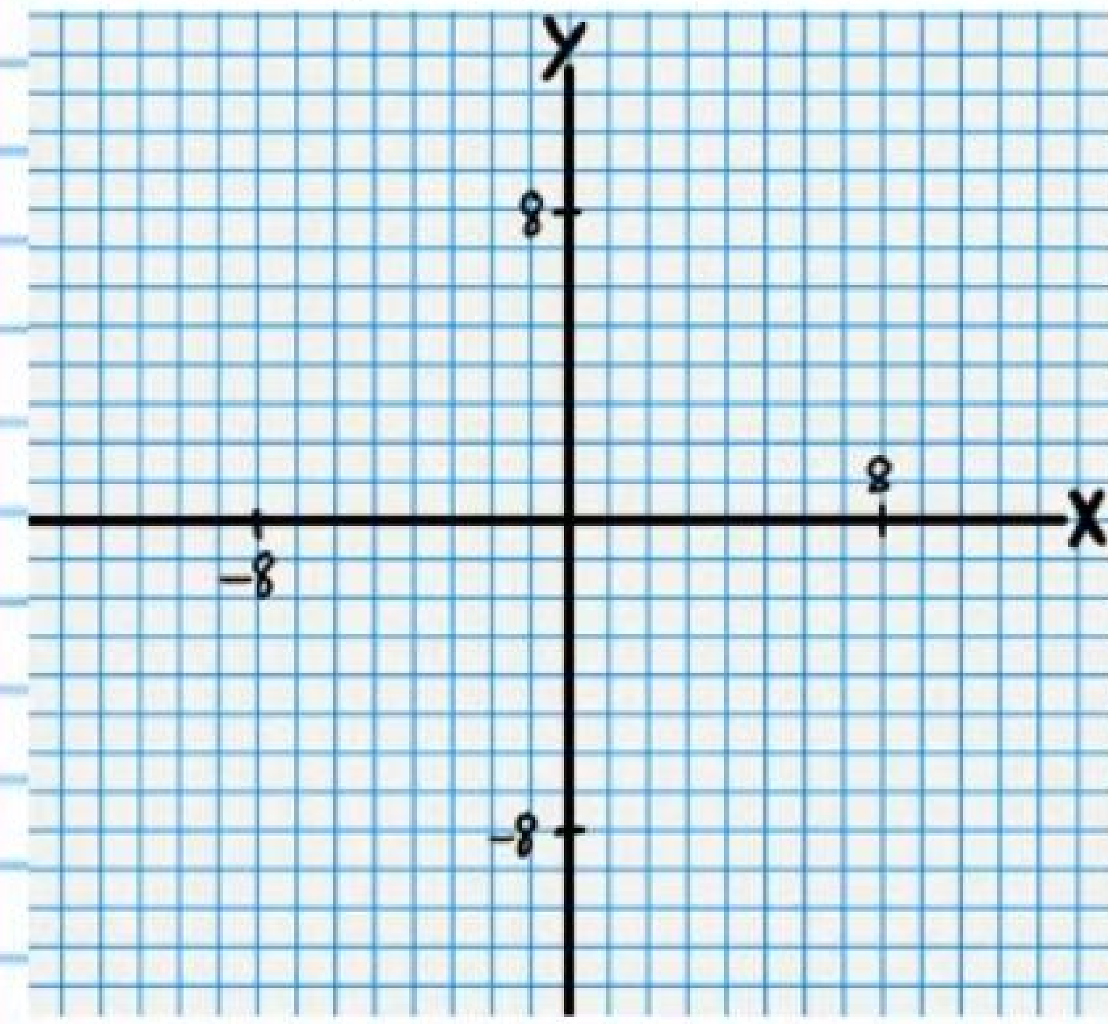
$-y = 3^x$



Transformation #2

Dilation

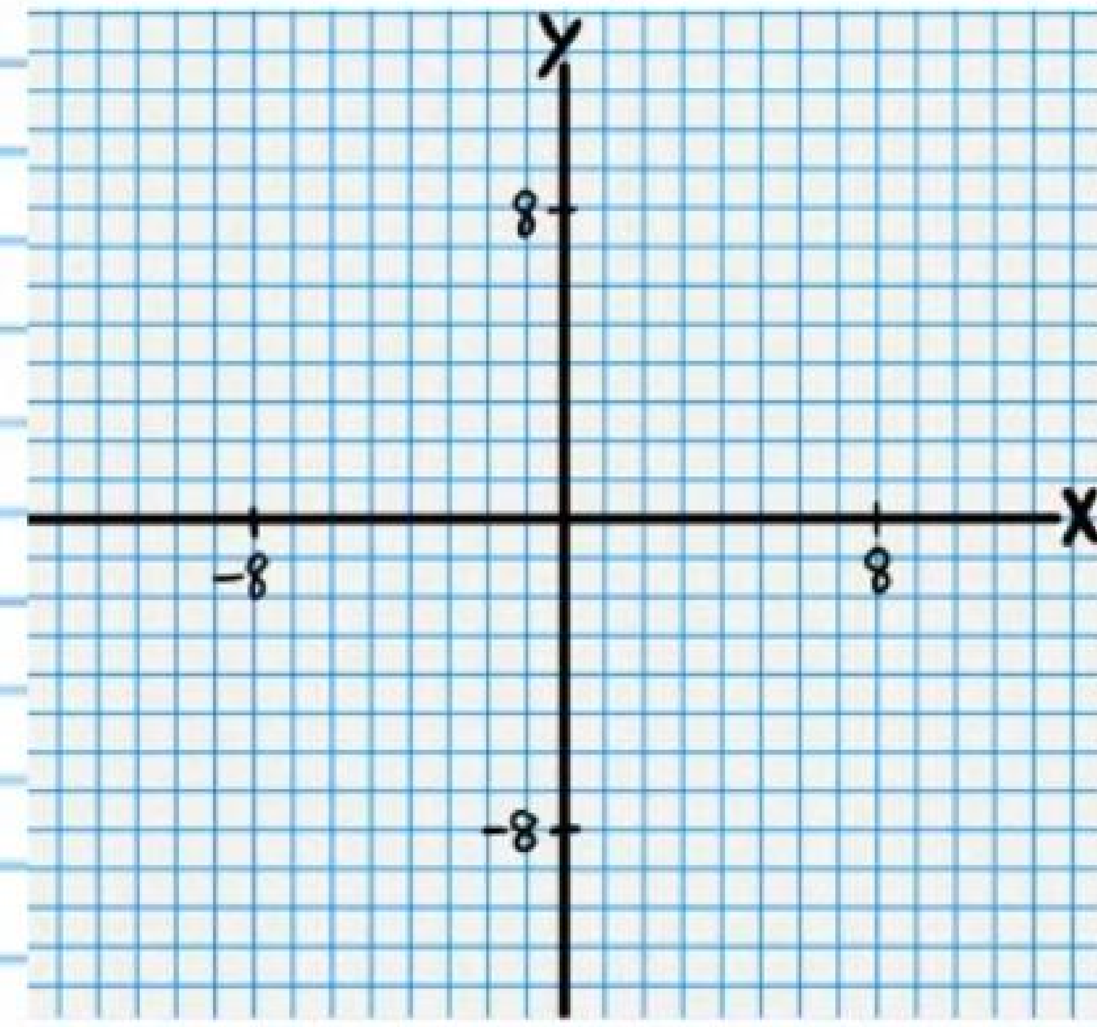
$-y = 3^{x/2}$



Final Transformation

Translation

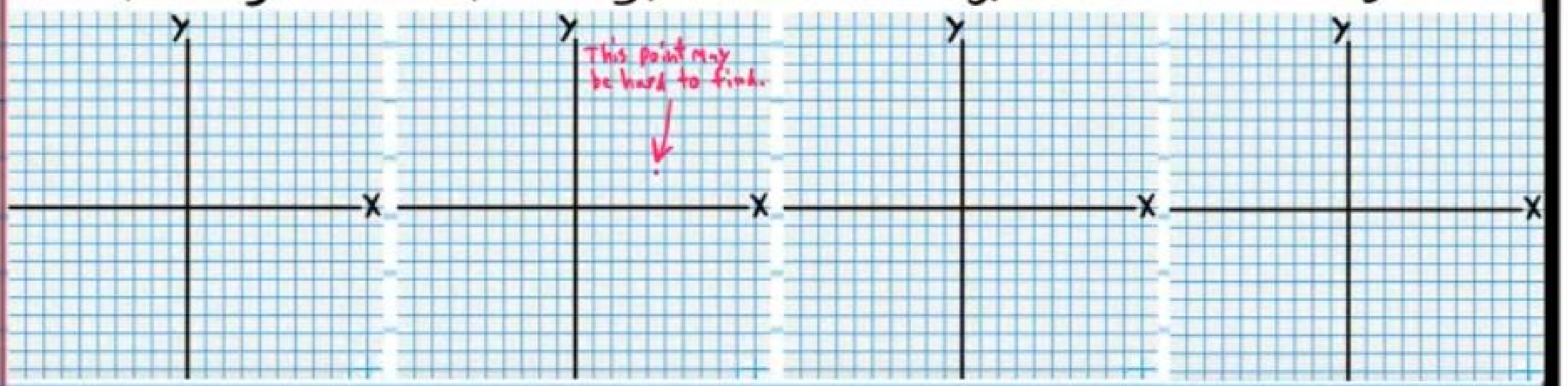
$-(y-2) = 3^{\frac{x-1}{2}}$



Logarithm Property III	Logarithm Property II	Logarithm Property IX
$\log_b b = 1$	$\log_b b^x = x$	$\log_b 1 = 0$

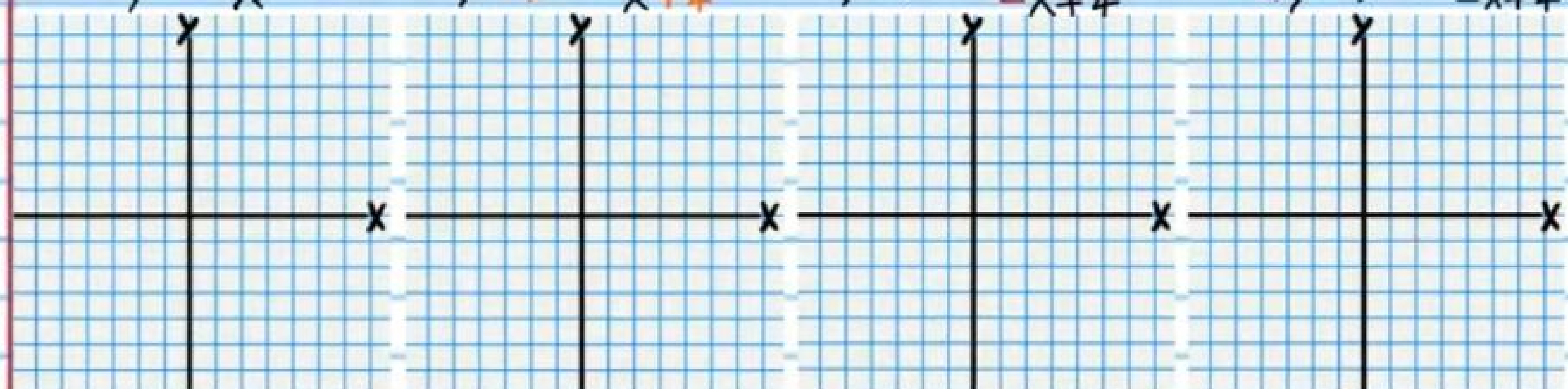
⑦ $-y+3=\log_3[2(x-2)]$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Dilation	Translation	Reflection
$y=\log_3 x$	$y=\log_3(2x)$	$y+3=\log_3[2(x-2)]$	$-y+3=\log_3[2(x-2)]$



⑧ $2y-3=\frac{1}{-x+4}$

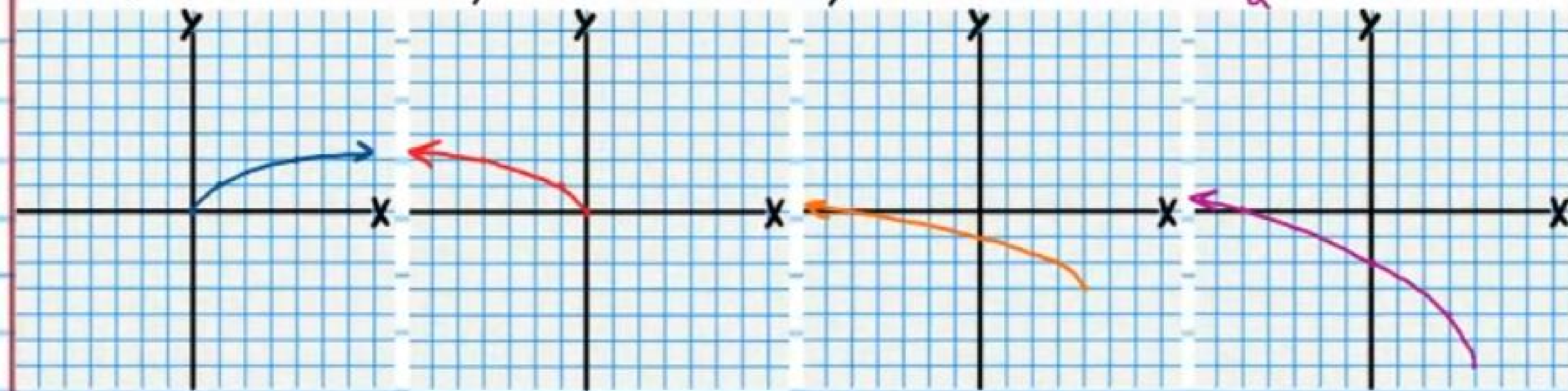
Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Translation	Reflection	Dilation
$y=\frac{1}{x}$	$y-3=\frac{1}{x+4}$	$y-3=\frac{1}{-x+4}$	$2y-3=\frac{1}{-x+4}$



Answers

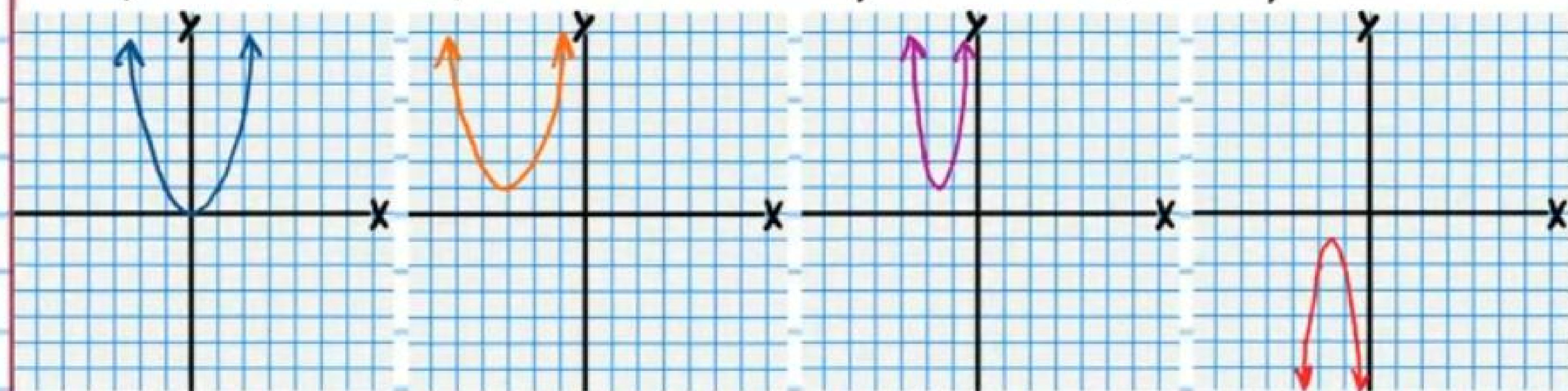
① $\frac{y}{2}+3=\sqrt{-(x-4)}$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Reflection	Translation	Dilation
$y=\sqrt{x}$	$y=\sqrt{-x}$	$y+3=\sqrt{-(x-4)}$	$\frac{y}{2}+3=\sqrt{-(x-4)}$



② $-y-1=(2x+3)^2$

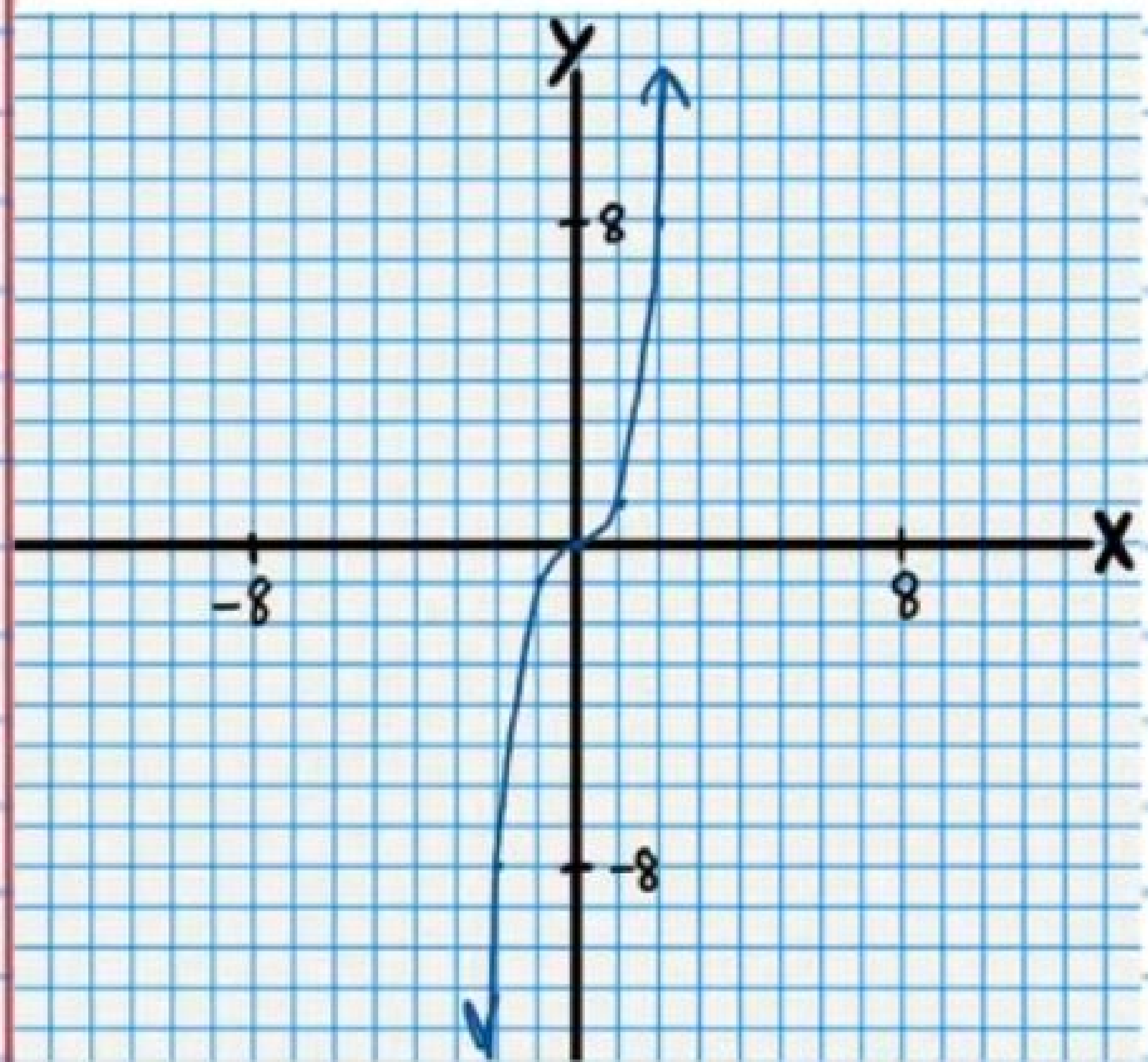
Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Translation	Dilation	Reflection
$y=x^2$	$y-1=(x+3)^2$	$y-1=(2x+3)^2$	$-y-1=(2x+3)^2$



③ $\frac{-(y-2)}{3}=(x-2)^3$

Parent Relation

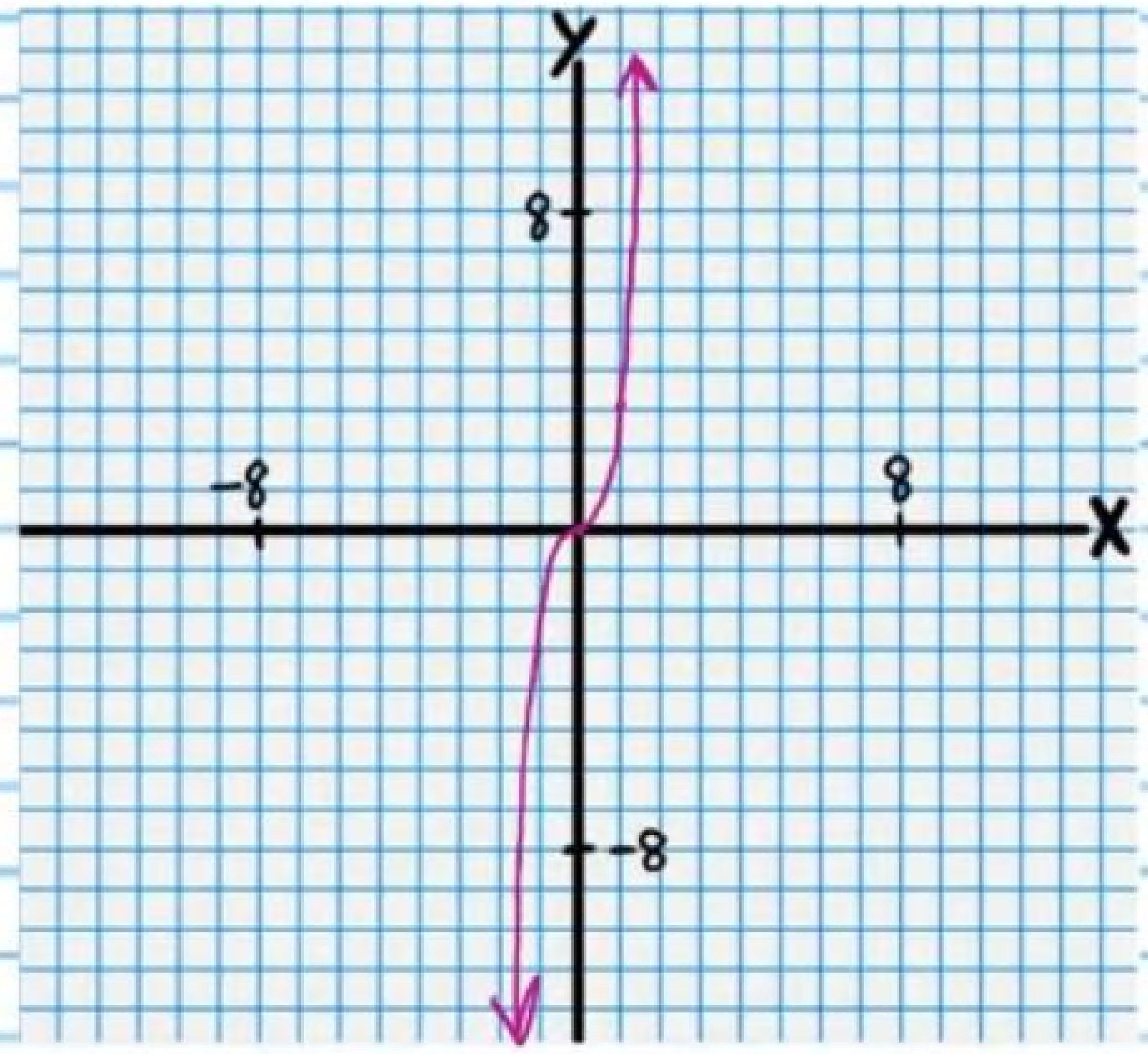
$y=x^3$



Transformation #1

Dilation

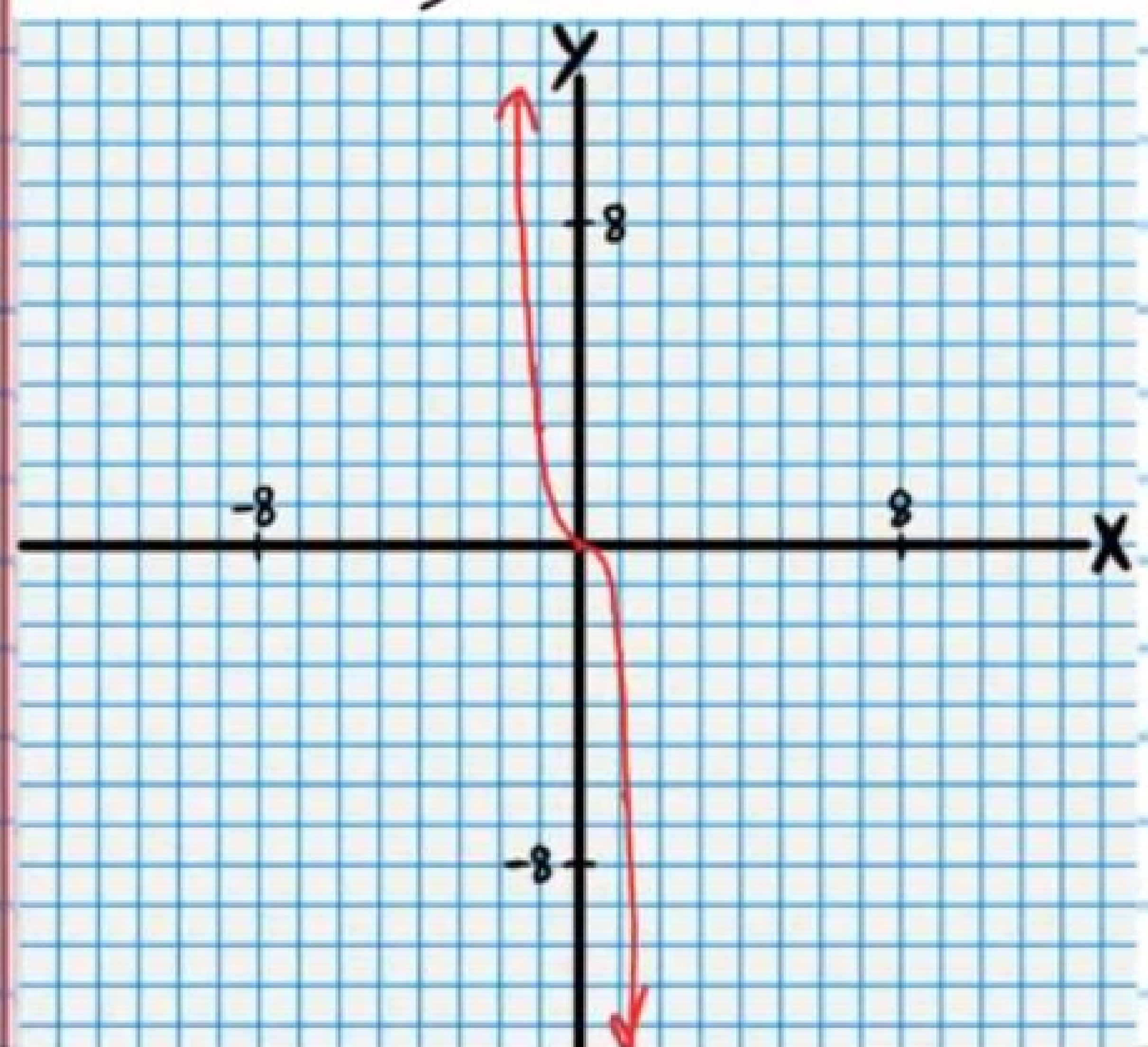
$\frac{y}{3}=x^3$



Transformation #2

Reflection

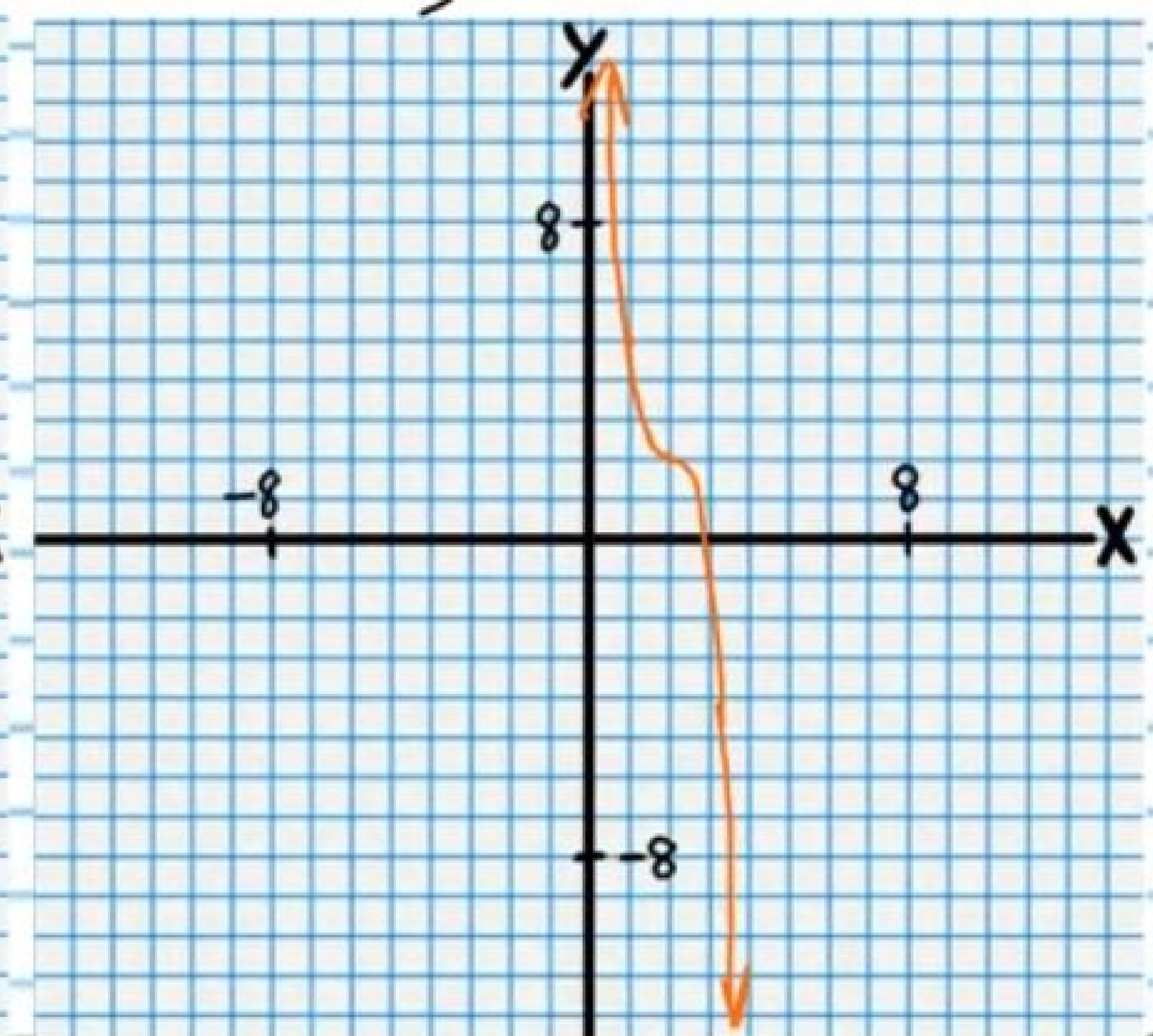
$-\frac{y}{3}=x^3$



Final Transformation

Translation

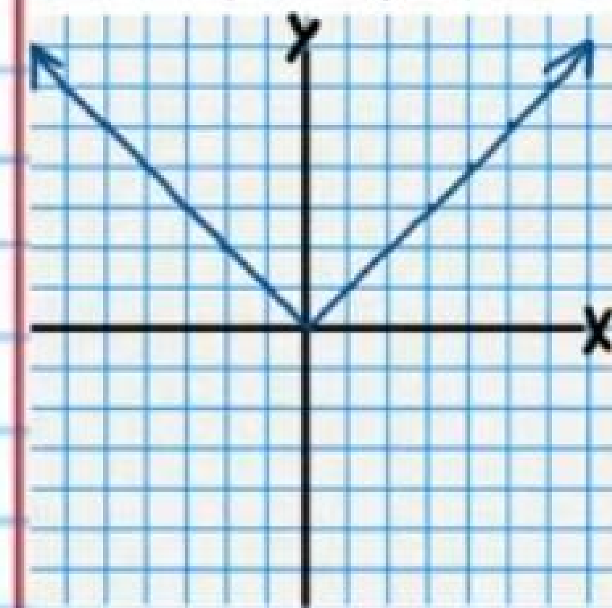
$\frac{-(y-2)}{3}=(x-2)^3$



④ $-(y+1)=\left|\frac{x}{3}+1\right|$

Parent Relation

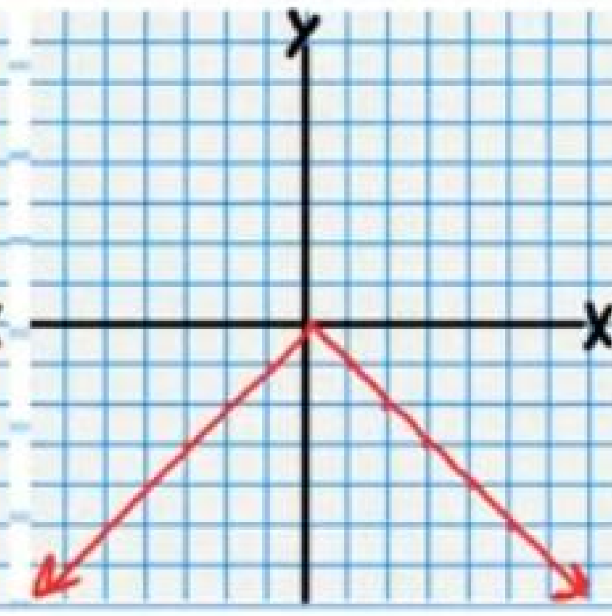
$y=|x|$



Transformation #1

Reflection

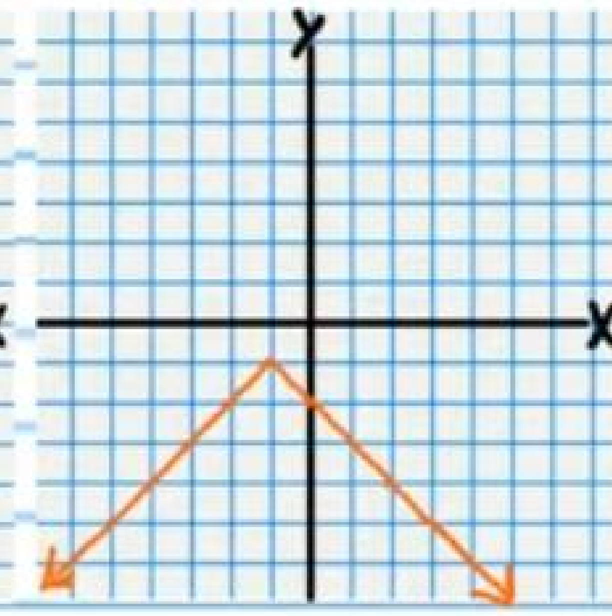
$-y=|x|$



Transformation #2

Translation

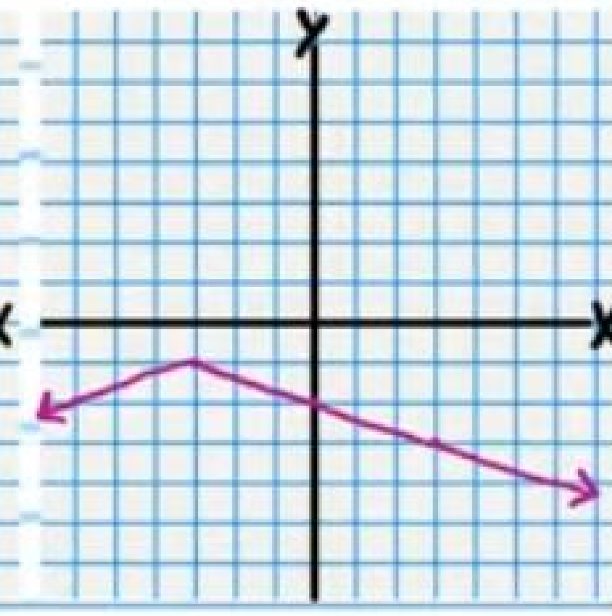
$-(y+1)=|x+1|$



Final Transformation

Dilation

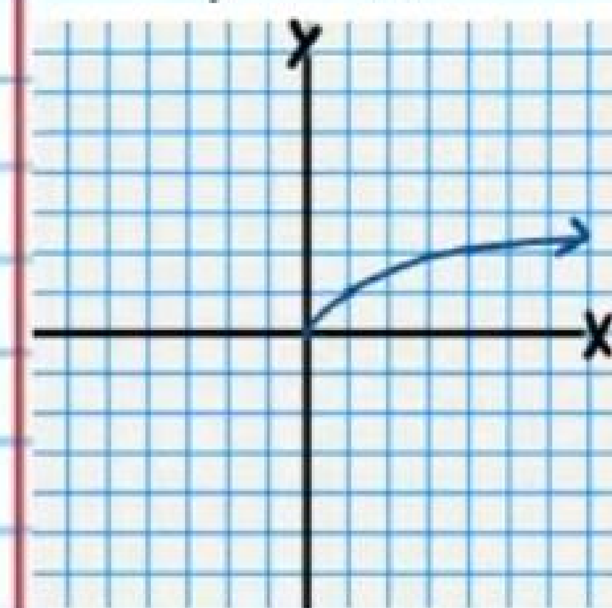
$-(y+1)=\left|\frac{x}{3}+1\right|$



⑤ $2y+3=\sqrt{-x+1}$

Parent Relation

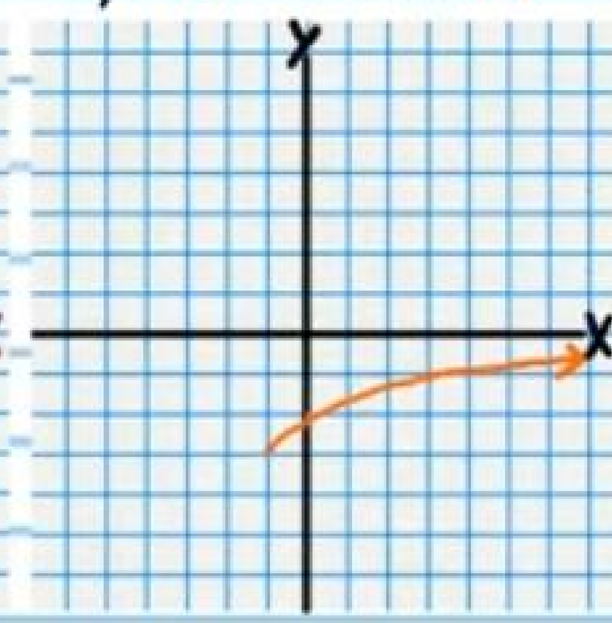
$y=\sqrt{x}$



Transformation #1

Translation

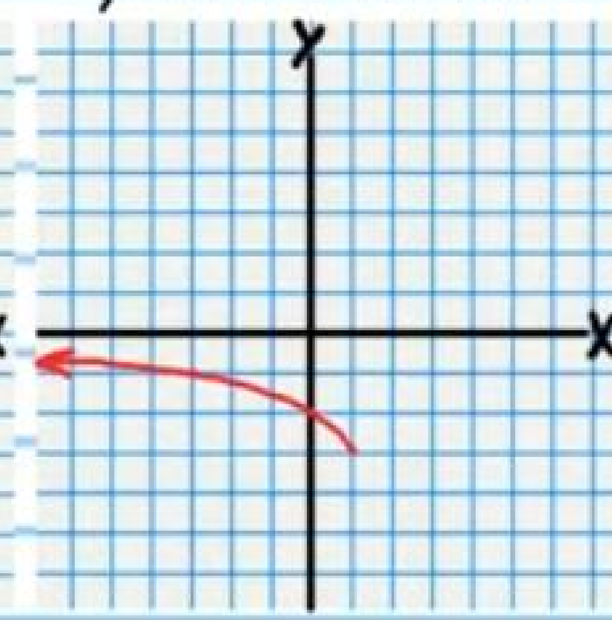
$y+3=\sqrt{x+1}$



Transformation #2

Reflection

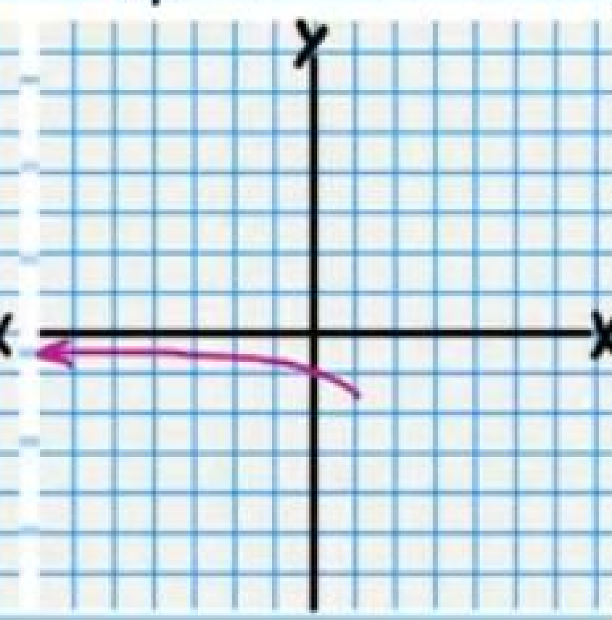
$y+3=\sqrt{-x+1}$



Final Transformation

Dilation

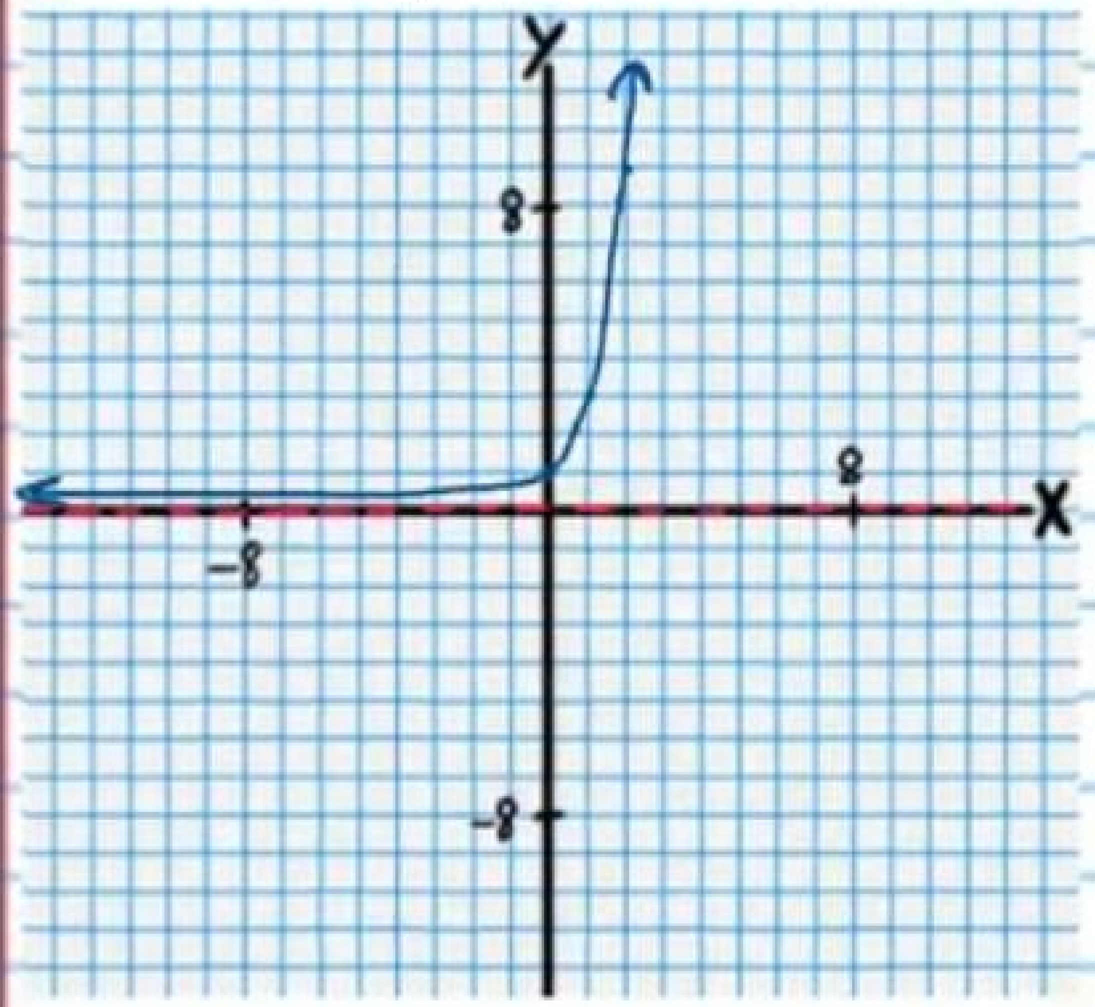
$2y+3=\sqrt{-x+1}$



⑥ $-(y-2) = 3^{\frac{x-1}{2}}$

Parent Relation

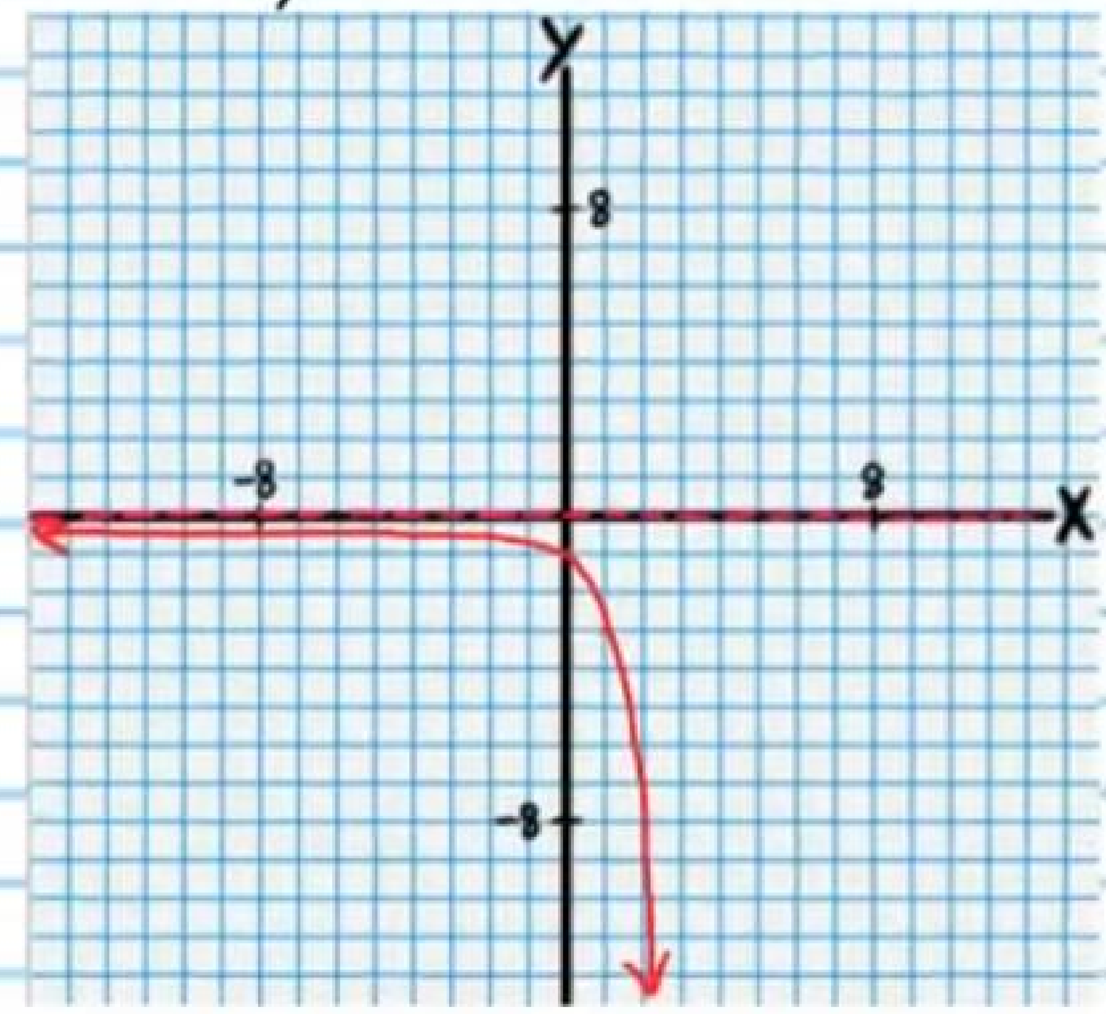
$y = 3^x$



Transformation #1

Reflection

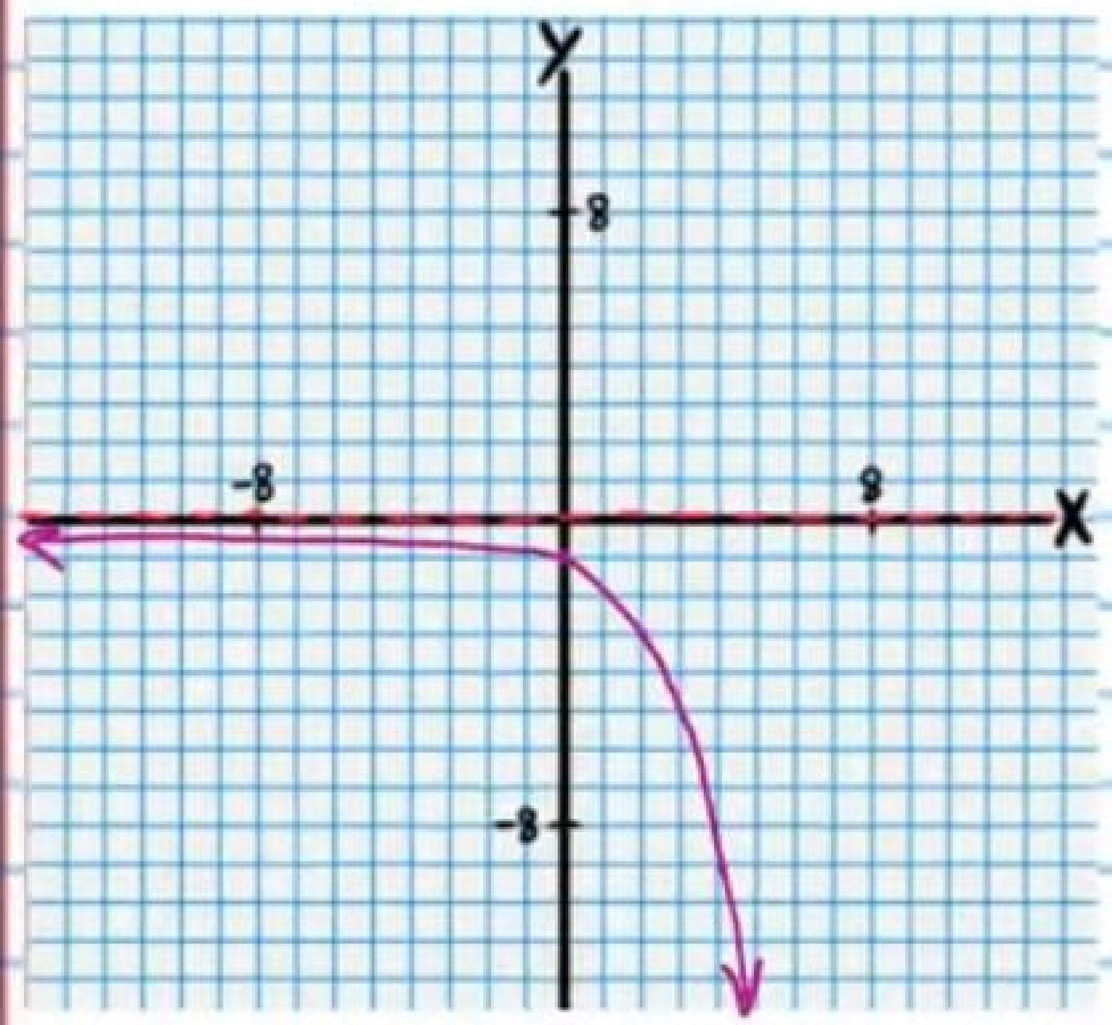
$-y = 3^x$



Transformation #2

Dilation

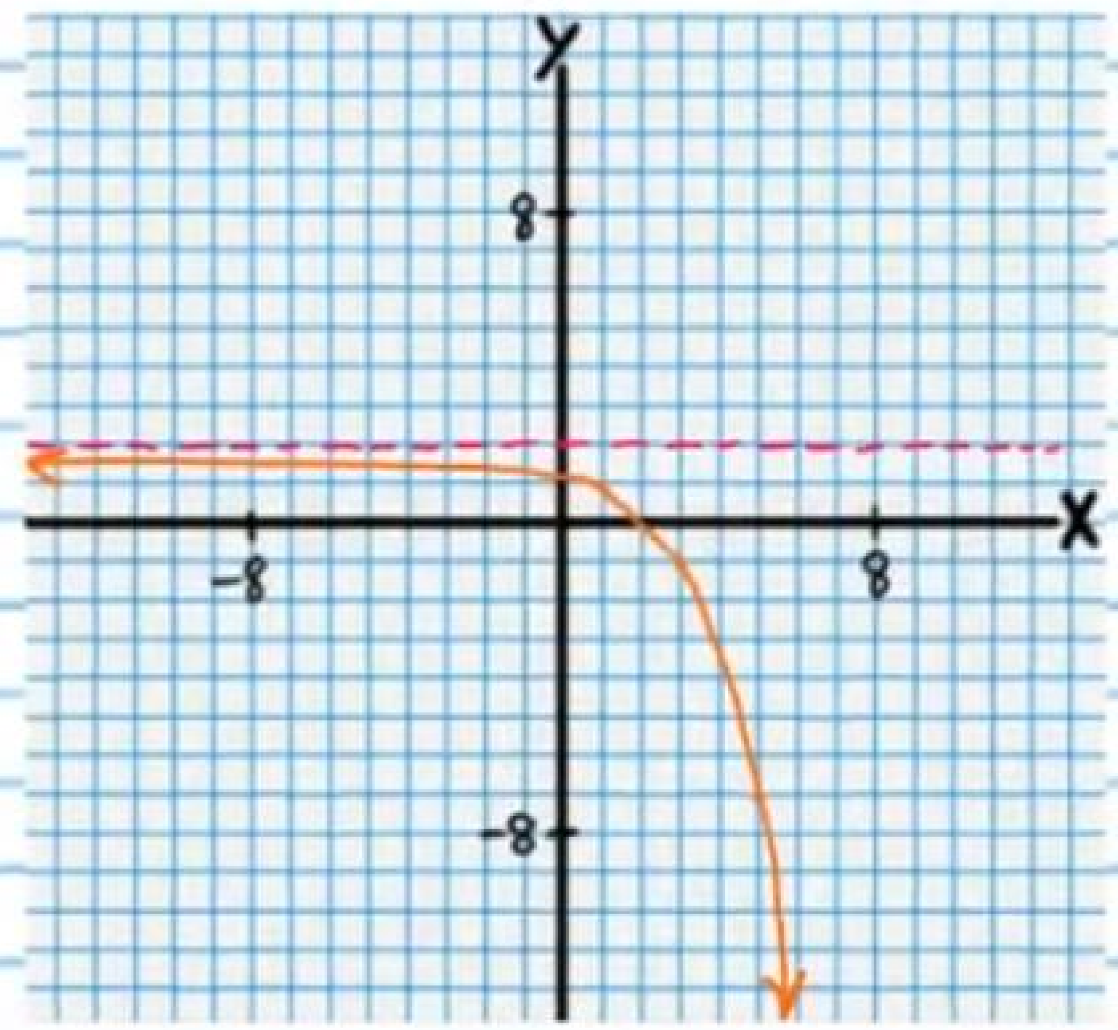
$-y = 3^{x/2}$



Final Transformation

Translation

$-(y-2) = 3^{\frac{x-1}{2}}$

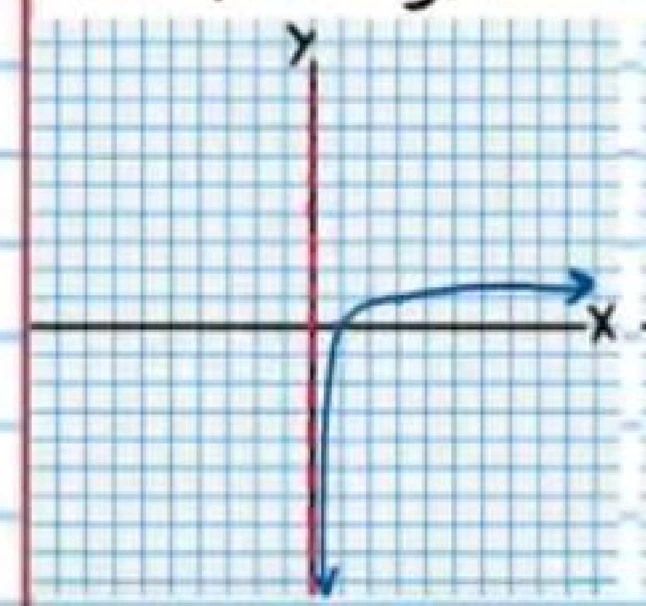


Logarithm Property III	Logarithm Property II	Logarithm Property IX
$\log_b b = 1$	$\log_b b^x = x$	$\log_b 1 = 0$

⑦ $-y+3 = \log_3[2(x-2)]$

Parent Relation

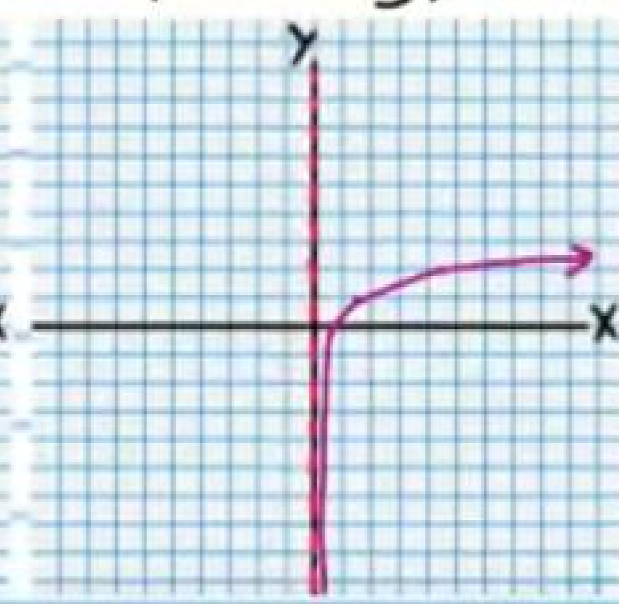
$y = \log_3 x$



Transformation #1

Dilation

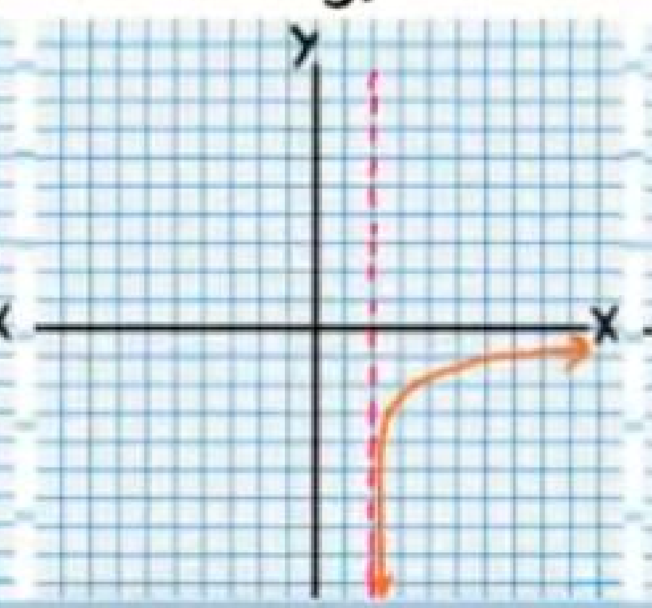
$y = \log_3(2x)$



Transformation #2

Translation

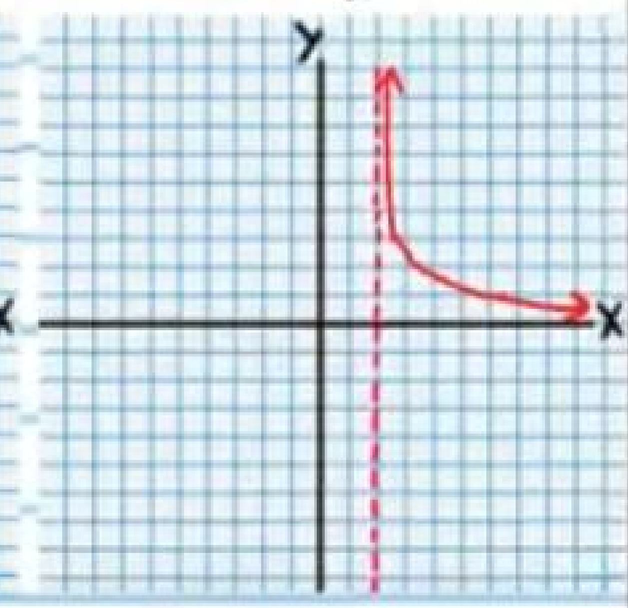
$y+3 = \log_3[2(x-2)]$



Final Transformation

Reflection

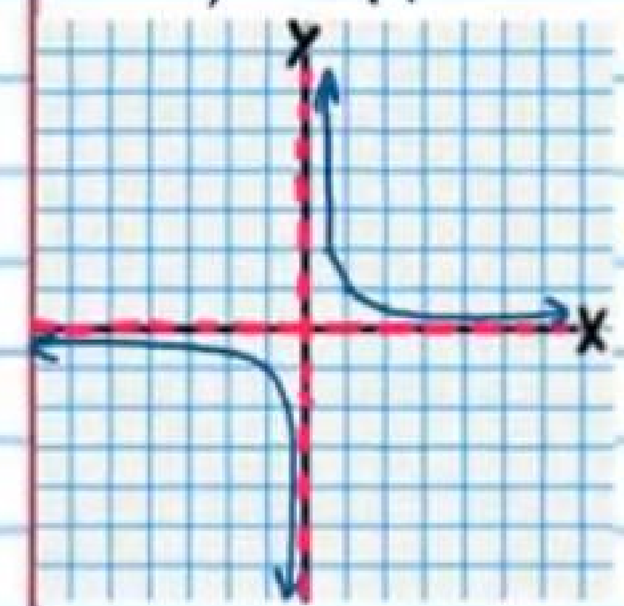
$-y+3 = \log_3[2(x-2)]$



⑧ $2y-3 = \frac{1}{-x+4}$

Parent Relation

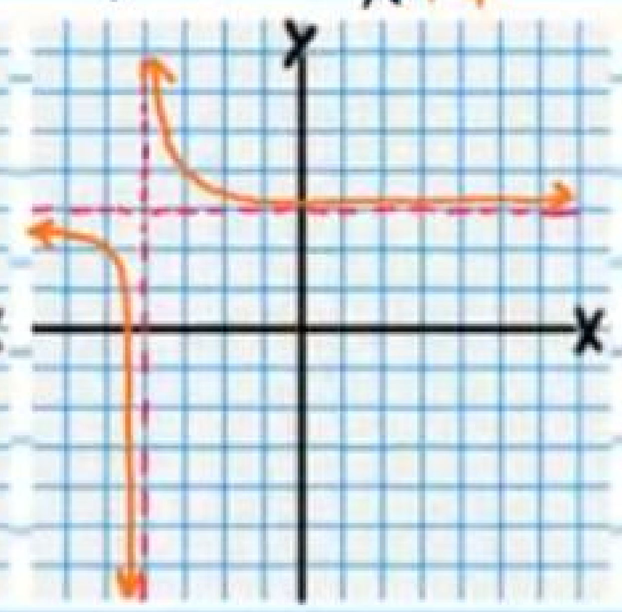
$y = \frac{1}{x}$



Transformation #1

Translation

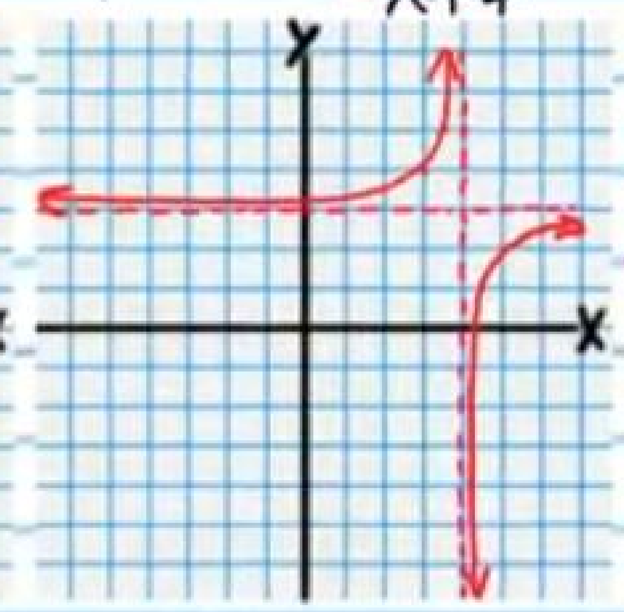
$y-3 = \frac{1}{x+4}$



Transformation #2

Reflection

$y-3 = \frac{1}{-x+4}$



Final Transformation

Dilation

$2y-3 = \frac{1}{-x+4}$